

AI ENGINEER CAREER ROADMAP

Personalized Strategic Plan · Student → AI Engineer

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TIMELINE: 12 MONTHS

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Current Position

Status: Student

Python

Web Development

Data Analysis

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Target Goal

Become an AI Engineer

Build ML/AI systems, deploy models, join an AI/ML team within 12 months

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Strategic Fit

Python + Data Analysis = **strong foundation**. Web Dev enables full-stack AI product building — a key differentiator vs. pure-ML candidates.

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Skill Gap Analysis

Skill Area	Priority	Current State	Required for AI Engineer
Machine Learning Fundamentals	HIGH	Not started	Supervised/unsupervised learning, model evaluation
Deep Learning (PyTorch/TF)	HIGH	Not started	Neural nets, CNNs, transformers
LLMs & GenAI / Prompt Eng.	HIGH	Not started	RAG, fine-tuning, agents, API integration
Math for ML (Linear Algebra, Stats)	MEDIUM	Partial (via Data Analysis)	Applied fluency for model design
MLOps & Deployment	MEDIUM	Not started	Docker, APIs, cloud (AWS/GCP), model serving
Python	STRONG	Existing skill	Extend to ML libraries (NumPy, Pandas, sklearn)
Data Analysis	STRONG	Existing skill	Directly transferable to feature engineering

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Recommended Learning Plan

Phase	Focus	Key Topics / Resources
Months 1-3	ML Foundations	Python for ML, NumPy/Pandas/sklearn, statistics, Andrew Ng's ML course, linear/logistic regression, decision trees
Months 4-6	Deep Learning	Neural networks, PyTorch, CNNs, RNNs, deep learning specialization, Kaggle competitions
Months 7-9	GenAI & LLMs	Transformers, Hugging Face, prompt engineering, RAG pipelines, LangChain, vector databases
Months 10-12	Deployment & Interview Prep	Docker, FastAPI, cloud deployment (AWS/GCP), system design for ML, mock interviews

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Suggested Projects

- M1-3:** Data analysis dashboard (Pandas + web dev skills combined)
- M4-6:** Image classifier (CNN) deployed as a web app
- M7-9:** RAG-based chatbot using your own document set
- M10-12:** End-to-end AI product: model + API + frontend + cloud deployment
- Bonus: Contribute to 1 open-source ML repo

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Networking Strategy

- Post weekly build-in-public updates on LinkedIn/X
- Join AI/ML Discord & Kaggle communities
- Attend 1 AI meetup/hackathon per quarter
- Connect with 5 AI engineers/month for informational chats
- Publish project write-ups on Medium/GitHub

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Monthly Milestones

Months 1-3	Months 4-6	Months 7-9	Months 10-12
- ML fundamentals done 1st project shipped - Kaggle account active	- Deep learning basics - CNN project deployed 150+ LinkedIn network	- RAG chatbot live 1 open-source PR merged - Resume/portfolio ready	- Capstone deployed 10+ interviews done - Job offer secured

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Immediate Next Actions (This Week)

1. Enroll in an ML foundations course (Andrew Ng / fast.ai)

2. Set up Python ML environment: NumPy, Pandas, scikit-learn, Jupyter

3. Create/update LinkedIn & GitHub to reflect "Aspiring AI Engineer" positioning

4. Pick and start Project 1 (data analysis dashboard) this week

Generated using Chain-of-Thought Reasoning